

% Constants

M = 1.5e-5; % m²/s (kinematic viscosity of air)

g = 9.81; % m/s² (acceleration due to gravity)

alpha = 2.11e-5; % m²/s (thermal diffusivity of air)

beta = 0.0034; % Coefficient of thermal expansion

L=100;

H=20;

T_plate= 65;

T_inf=25;

% Grid parameters

Nx = 10; % Number of grid points along x-axis

Ny = 10; % Number of grid points along y-axis

dx = L / (Nx - 1); % Grid spacing in x-direction

dy = H / (Ny - 1); % Grid spacing in y-direction

% Initialize arrays for velocity components (v) and temperature (T)

v = zeros(Ny, Nx);

T = zeros(Ny, Nx);

% Boundary conditions

v(:, 1) = 0; % No-slip condition at the plate

T(:, 1) = T_plate;

% Iterative solver

max_iter = 1000;

tolerance = 1e-6;

for iter = 1:max_iter

 v_old = v;

```
T_old = T;
```

```
% Update velocity components using the discretized momentum equation
```

```
for i = 2:Nx-1
```

```
    for j = 2:Ny-1
```

```
        v(j, i) = (M * ((v_old(j, i+1) - 2*v_old(j, i) + v_old(j, i-1)) / dy^2) + ...
```

```
            g * beta * (T_old(j, i) - T_inf)) * dy^2 + v_old(j, i);
```

```
    end
```

```
end
```

```
% Update temperature using the discretized energy equation
```

```
for i = 2:Nx-1
```

```
    for j = 2:Ny-1
```

```
        T(j, i) = (alpha * ((T_old(j, i+1) - 2*T_old(j, i) + T_old(j, i-1)) / dy^2)) * dy^2 +  
T_old(j, i);
```

```
    end
```

```
end
```

```
% Check for convergence
```

```
if max(max(abs(v - v_old))) < tolerance && max(max(abs(T - T_old))) < tolerance
```

```
    fprintf('Converged at iteration: %d\n', iter);
```

```
    break;
```

```
end
```

```
end
```

```
% Plotting
```

```
[X, Y] = meshgrid(linspace(0, L, Nx), linspace(0, H, Ny));
```

```
% Plotting temperature field and streamlines
```

```
figure;  
contourf(X, Y, T, 'LineColor', 'none');  
colorbar;  
hold on;  
quiver(X, Y, u, v);  
title('Temperature Distribution and Streamlines');  
xlabel('X');  
ylabel('Y');  
hold off;
```